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#include <iostream>
#include <cmath>
using namespace std;
double f(double x) {
    return x * x;
}
double gaussQuadrature(double a, double b) {
    double weights[2] = {1.0, 1.0};
    double points[2] = {-1.0 / sqrt(3), 1.0 / sqrt(3)};
    double result = 0.0;
    double midpoint = (b + a) / 2.0;
    double half_width = (b - a) / 2.0;
    for (int i = 0; i < 2; i++) {
        result += weights[i] * f(midpoint + half_width * points[i]);
    }
    return result * half_width;
}
int main() {
    double a, b;
    cout << "Enter the lower limit (a): ";
    cin >> a;
    cout << "Enter the upper limit (b): ";
    cin >> b;
    double result = gaussQuadrature(a, b);
    cout << "The approximate integral from " << a << " to " << b << " is: " << result << endl;
    return 0;
}

```